

# Darren Zheng

 [Personal Website](#)

 [dzheng24@sas.upenn.edu](mailto:dzheng24@sas.upenn.edu)

 [LinkedIn](#)

## EDUCATION

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### University of Pennsylvania

Aug 2021 – May 2025

*B.A. in Mathematics, Minor in Statistics & Data Science, Ben Franklin Scholar*

*B.A. GPA: 3.51/4.0*

- Relevant Coursework: Data Science in Python, Statistical Computing with R, Optimization, Computer Algorithms, Corporate Finance, Probability Theory, Stochastic Processes, Statistical Inference, Partial Differential Equations

## PROJECTS

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### Undergraduate Honor's Thesis

June 2024 – May 2025

- Constructed intrinsic Morse functions on Grassmannian manifolds  $G_k(\mathbb{R}^4)$  to compute singular homology groups

### Directed Reading Program

Aug 2022 – May 2025

- Engaged in independent study across 6 semesters, covering knot theory, generating functions, Brownian motion, stochastic calculus, additive combinatorics, Falconer's conjecture, and mapping class groups
- Prepared expository write-ups and delivered presentations to the Mathematics Department, translating abstract concepts to a diverse audience of students and professors

### Detecting Warning Signals of Financial Collapse using Topological Data Analysis

Jan 2024 – Apr 2024

- Applied persistent homology techniques to time-series data to develop early warning signals for market crashes
- Implemented Python pipeline (`scikit-tda`, `risper`, `persim`) to transform historic stock index data into persistence landscapes for sliding window analysis
- Conducted statistical validation using Mann-Kendall test to analyze significance of methodology

## EXPERIENCE

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### Private Competitive Mathematics Coaching | Tutor

May 2025 – Present

- Guide 14 students with individualized training for math contests, such as AIME, USAMO, and MathCounts
- Prepare personalized problem sets & mock exams to target technical gaps based on past student performance

### University of Pennsylvania Math Department | Teaching Assistant

Aug 2022 – May 2025

- Developed and delivered weekly recitations for various proof-based calculus and linear algebra courses
- Graded homework/exams, providing feedback on proof-writing; Received 2 Departmental teaching awards

### Ross Mathematics Program | Administrator

June 2024 – July 2024

- Managed operations for a 6-week residential summer math camp for  $\sim 80$  high schoolers and  $\sim 20$  instructional staff
- Scheduled social and educational programming to minimize time conflicts, coordinating with the host university
- Presented some mathematical talks, involving formal logic, algebraic and point-set topology to advanced students

### AwesomeMath Summer Program | Teaching Assistant

June - Aug 2022/2023/2025

- Instructed six 3-week-long courses of  $\sim 15$ -30 participants, facilitating problem sessions and active learning
- Taught primarily number theory, combinatorics, and abstract algebra

## ACTIVITIES

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### Penn Math Contest Club | Co-founder & President

Aug 2021 – May 2025

- Established a new student club that organizes an on-campus high-school mathematics contest with original problems
- Planned event logistics, coordinating with campus facilities & training  $\sim 30$  club members/volunteers; Secured corporate sponsorships to offset operational expenses. Created material for future leadership
- Grew from  $\sim 40$  online participants (COVID) to  $\sim 380$  participants in Spring 2025 through outreach to schools

### Penn Undergraduate Math Society | President

Jan 2024 – May 2025

- Led Penn's undergraduate math club, directing a board of  $\sim 10$  to support math-adjacent programming
- Organized various events ( $\sim 13$  per semester), including major/course advising, research talks, integration bee, merch sale, faculty lunches, DRP x PUMS collaboration, and industry panels

### Science Olympiad at Penn | Test Writer & Event Supervisor

Aug 2021 - May 2025

- Authored 7 state-level tests over 4 years and designed practical lab components, specializing in chronology, astronomy, cryptography, optics and Fermi estimates
- Utilized automated rubrics and guidelines for volunteers to ensure consistent scoring across 200+ participants

### Student Activities Council | Executive Board

Feb 2023 – Dec 2024

- Oversaw strategic allocation of an  $\sim 1.7$  million annual budget to over 300 student groups, determining budgets and representing their interests to the University administration
- Spearheaded changes to club debt policies and interviews for the new group recognition process

## TECHNICAL SKILLS

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**Languages:** Python, Java, SQL, Go, JavaScript

**Tools:** NumPy, Pandas,  $\text{\LaTeX}$ , Excel VBA, HTML/CSS, Git